

Chapter 1

INTRODUCTION

Plastic waste has become one of the major environmental challenges due to the increasing use of plastic products and improper disposal methods. Large amounts of plastic waste end up in landfills, water bodies, and public areas, causing pollution and harming ecosystems. To address this issue, the project “Smart Plastic Recycling: A Reward-Based Solution” aims to encourage responsible plastic waste management through a simple and user-friendly digital platform.

The proposed system allows users to register, schedule plastic waste pickups, and hand over recyclable plastic materials from their doorstep. In return, users receive reward points that motivate active participation in recycling activities. The platform is developed using modern web technologies such as React.js, Node.js, and MongoDB to provide efficient data management and smooth user interaction

1.1 Brief History

Traditional plastic waste management systems mainly relied on manual collection methods and public recycling bins. People had to separately collect and transport plastic waste to recycling centers, which reduced public participation due to inconvenience and lack of awareness. These systems also lacked proper monitoring, motivation, and efficient waste segregation methods. As a result, a large amount of plastic waste was disposed of improperly, causing pollution in landfills, rivers, and oceans. The absence of reward mechanisms and digital support further limited the effectiveness of recycling activities.

1.2 Modern System

The modern system, “Smart Plastic Recycling: A Reward-Based Solution,” uses digital technology to improve recycling efficiency and encourage community participation. The platform allows users to register, schedule doorstep plastic waste pickups, and earn reward points for recycling activities. Technologies such as React.js, Node.js, and MongoDB are used for smooth operation and secure data

management. Features like automated scheduling, SMS notifications, and reward systems make recycling more convenient, accessible, and environmentally sustainable

Chapter 2

Problem Statement

2.1 Description

Plastic waste has become a major environmental problem due to excessive usage and improper disposal practices. Traditional recycling systems are often inconvenient and fail to encourage active public participation. Many people do not separate recyclable plastic waste because of lack of awareness, limited access to recycling centers, and absence of motivation. As a result, large amounts of plastic waste accumulate in landfills, water bodies, and public places, causing environmental pollution and health hazards.

The project, “Smart Plastic Recycling: A Reward-Based Solution,” is designed to simplify plastic waste management through a digital platform. The system allows users to register, schedule plastic waste pickups, and hand over recyclable materials from their doorstep. Users receive reward points for their contribution, which encourages regular recycling habits and promotes environmental sustainability.

2.2 Challenge Statement

The main challenge is to improve public participation in plastic recycling by creating a convenient and motivating waste management system. Existing recycling methods lack accessibility, awareness, and reward mechanisms, leading to low recycling rates and increased environmental pollution. The project aims to provide an efficient, user-friendly, and reward-based platform that encourages responsible plastic waste disposal and supports sustainable recycling practices.

Chapter 3

3.1 Design Thinking Process

- a) **Empathize** : Discussions with students and local residents identified problems such as lack of awareness, inconvenience in recycling, and absence of motivation for plastic waste disposal
- b) **Define** : The main needs identified were doorstep waste collection, simple scheduling, and a reward system to encourage recycling participation.
- c) **Ideate** : Different ideas such as smart bins and collection systems were discussed. The final solution selected was a reward-based digital recycling platform.
- d) **Prototype** : A web application prototype was developed using React.js, Node.js, and MongoDB with user registration and pickup scheduling features.
- e) **Test** : The system was tested for usability and convenience, showing improved participation in recycling activities.

3.2 Methodology



3.3 Prototype Description

The prototype is a web-based smart plastic recycling platform named EcoCollect developed to simplify plastic waste management and encourage recycling participation. The system allows users to register, log in securely, schedule plastic waste pickups, and manage recycling requests through a digital dashboard. The platform provides a simple and user-friendly interface for efficient waste collection management.

The prototype was developed using frontend technologies such as HTML, CSS, and JavaScript, while Firebase was used for backend services including authentication and database management. Firestore Database is used to store user and pickup request details securely. The system also supports email verification and protected user access to improve security and reliability.

3.3.1 Materials Used

- a) **HTML** – Structure and design of web pages
- b) **CSS** – Styling and responsive user interface
- c) **JavaScript** – Interactive functionalities and validation
- d) **Firebase Authentication** – Secure login and user verification
- e) **Firestore Database** – Storage of user and pickup information
- f) **Firebase Hosting** – Deployment and online accessibility
- g) **Dashboard Interface** – Management of pickup requests and recycling activities

3.3.2 System Diagram

The Smart Plastic Recycling System follows a web-based client–server architecture. The frontend is developed using HTML, CSS, and JavaScript to

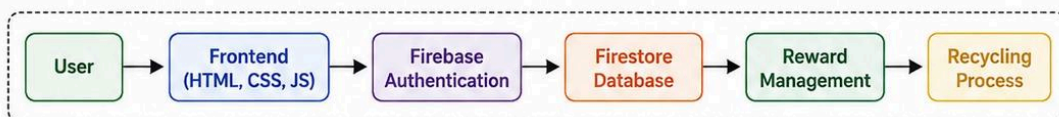
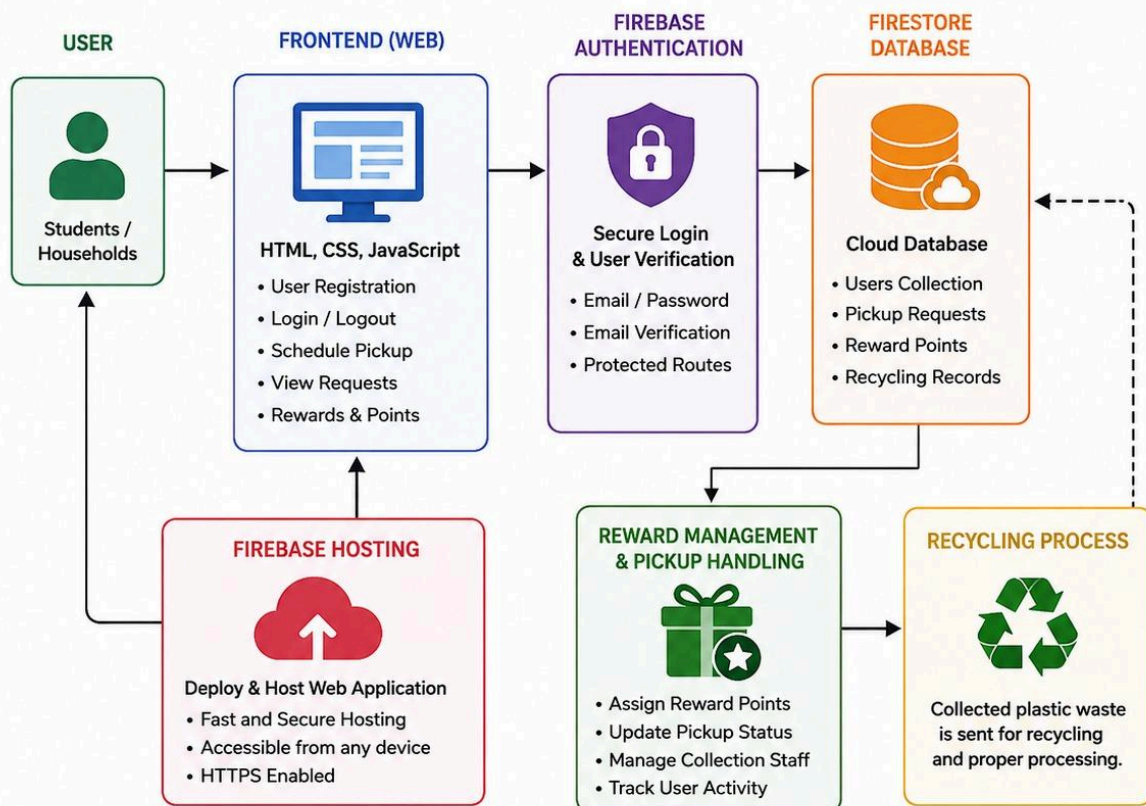
provide an interactive user interface for registration, login, and pickup scheduling. Firebase Authentication is used to securely manage user login and verification.

The system uses Firestore Database to store user information, pickup requests, and reward details. All user requests are processed through the application interface and securely stored in the cloud database. Firebase Hosting is used to deploy the application online, making the platform accessible from different devices.

The architecture enables smooth communication between users and the database while ensuring secure access, efficient data management, and reliable platform performance.

3.3.2 System Diagram

Software Architecture / Product Architecture



DESCRIPTION

The system allows users to register, login and schedule plastic waste pickup. Authentication is handled securely by Firebase Authentication. All data including users, pickup requests and rewards are stored in Firestore Database. The system manages reward points and pickup operations, and finally the collected plastic waste is sent for recycling. The application is hosted on Firebase Hosting for secure and reliable access.

Chapter 4

Implementation

```
firebase-config.js

01 // =====
02 // ECOCOLLECT – Firebase Configuration
03 // =====
04 // 1. Go to https://console.firebase.google.com
05 // 2. Create a new project called "ecocollect"
06 // 3. Click the </> (Web) icon to add a web app
07 // 4. Copy your config values below
08 // =====
09
10 const firebaseConfig = {
11   apiKey:      "YOUR_API_KEY",
12   authDomain:  "YOUR_PROJECT_ID.firebaseio.com",
13   projectId:   "YOUR_PROJECT_ID",
14   storageBucket: "YOUR_PROJECT_ID.appspot.com",
15   messagingSenderId: "YOUR_SENDER_ID",
16   appId:      "YOUR_APP_ID"
17 };
18
19 // Initialize Firebase
20 firebase.initializeApp(firebaseConfig);
21 const auth = firebase.auth();
22 const db   = firebase.firestore();
23
24 // ——— Firestore Security Rules (paste in Firebase Console → Firestore → Rules) —
25 /*
26 rules_version = '2';
27 service cloud.firestore {
28   match /databases/{database}/documents {
29     match /users/{userId} {
30       allow read, write: if request.auth != null && request.auth.uid == userId;
31     }
32     match /pickupRequests/{requestId} {
33       allow read, write: if request.auth != null
34         && request.auth.uid == resource.data.userId;
35       allow create: if request.auth != null;
36     }
37     match /collectors/{collectorId} {
38       allow read: if request.auth != null;
39     }
40   }
41 }
42 */
43
44 // ——— Seed a demo collector (run once in browser console) —————
45 /*
```


dashboard.html

```
01 <!DOCTYPE html>
02 <html lang="en">
03 <head>
04 <meta charset="UTF-8">
05 <meta name="viewport" content="width=device-width, initial-scale=1.0">
06 <title>EcoCollect – Dashboard</title>
07 <link href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:opsz,wght@9..40,300;9..40,400;9..40,500;9..40,600;9..40,700;9..40,800;9..40,900&display=block">
08
09 <!-- Firebase SDKs -->
10 <script src="https://www.gstatic.com/firebasejs/10.7.1/firebase-app-compat.js"></script>
11 <script src="https://www.gstatic.com/firebasejs/10.7.1/firebase-auth-compat.js"></script>
12 <script src="https://www.gstatic.com/firebasejs/10.7.1/firebase-firestore-compat.js"></script>
13 <script src="firebase-config.js"></script>
14
15 <style>
16 :root {
17   --bg:#0f1a14; --surface:#172117; --surface2:#1e2d20; --surface3:#253228;
18   --border:rgba(255,255,255,.07); --border2:rgba(255,255,255,.12);
19   --green-500:#40916c; --green-400:#52b788; --green-300:#74c69d; --green-100:#d8f3dc;
20   --amber:#e76f51; --amber-lt:#f4a261;
21   --text:#e8f5e9; --text-mid:rgba(232,245,233,.7); --text-soft:rgba(232,245,233,.4);
22   --sidebar-w:240px; --r:8px; --r-lg:14px; --r-xl:20px;
23   --red:#ef5350; --blue:#42a5f5; --yellow:#ffcda28;
24 }
25 *, ::before, ::after { box-sizing:border-box; margin:0; padding:0; }
26 html,body { height:100%; font-family:'DM Sans',sans-serif; background:var(--bg); color:var(--text); overflow:hidden; }
27
28 /* — LAYOUT — */
29 .app { display:flex; height:100vh; }
30
31 /* — SIDEBAR — */
32 .sidebar {
33   width:var(--sidebar-w); min-height:100vh; background:var(--surface);
34   border-right:1px solid var(--border); display:flex; flex-direction:column;
35   padding:24px 0; flex-shrink:0; overflow-y:auto;
36 }
37 .sidebar-brand { display:flex; align-items:center; gap:10px; padding:0 20px 28px; border-bottom:1px solid var(--border); margin-bottom:10px; }
38 .sidebar-brand-icon { width:36px; height:36px; background:var(--green-500); border-radius:10px; display:grid; place-items:center; }
39 .sidebar-brand-name { font-family:'Syne',sans-serif; font-weight:800; font-size:1.1rem; color:var(--text); }
40
41 .nav-section { padding:8px 12px; margin-top:4px; }
42 .nav-section-label { font-size:.68rem; font-weight:700; letter-spacing:1.5px; text-transform:uppercase; color:var(--text-soft); }
43 .nav-item {
44   display:flex; align-items:center; gap:12px; padding:10px 12px;
45   border-radius:var(--r-lg); cursor:pointer; transition:all .2s;
46 }
```

auth.html

```
01 <!DOCTYPE html>
02 <html lang="en">
03 <head>
04 <meta charset="UTF-8">
05 <meta name="viewport" content="width=device-width, initial-scale=1.0">
06 <title>EcoCollect – Sign In / Register</title>
07 <link href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:opsz,wght@9..40,300;9..40,400;9..40,500;9..40,600;9..40,700;9..40,800;9..40,900;9..40,1000" rel="stylesheet">
08
09 <!-- Firebase SDKs -->
10 <script src="https://www.gstatic.com/firebasejs/10.7.1/firebase-app-compat.js"></script>
11 <script src="https://www.gstatic.com/firebasejs/10.7.1/firebase-auth-compat.js"></script>
12 <script src="https://www.gstatic.com/firebasejs/10.7.1/firebase-firestore-compat.js"></script>
13 <script src="firebase-config.js"></script>
14
15 <style>
16   :root {
17     --green-900:#0d2b1a; --green-800:#1b4332; --green-700:#2d6a4f;
18     --green-500:#40916c; --green-300:#74c69d; --green-100:#d8f3dc;
19     --amber:#e76f51; --amber-lt:#f4a261;
20     --cream:#fafaf7; --white:#ffffff;
21     --text-dark:#1a2e1f; --text-mid:#3a5a40; --text-soft:#6b8f71;
22     --error:#e53935; --error-bg:#ffebee;
23     --r:8px; --r-lg:16px; --r-xl:24px;
24   }
25   *, ::before, ::after { box-sizing:border-box; margin:0; padding:0; }
26   body { font-family:'DM Sans',sans-serif; min-height:100vh; display:flex; background:var(--cream); }
27
28   /* — LEFT PANEL — */
29   .left {
30     width:42%; min-height:100vh;
31     background:var(--green-800);
32     display:flex; flex-direction:column; justify-content:center;
33     padding:64px 56px; position:relative; overflow:hidden;
34   }
35   .left::before {
36     content:''; position:absolute; inset:0;
37     background:radial-gradient(ellipse 80% 60% at 20% 80%, rgba(116,198,157,.15) 0%, transparent 60%),
38       radial-gradient(ellipse 60% 40% at 80% 20%, rgba(231,111,81,.1) 0%, transparent 50%);
39   }
40   .left-content { position:relative; z-index:1; }
41   .brand { display:flex; align-items:center; gap:12px; margin-bottom:64px; text-decoration:none; }
42   .brand-icon { width:44px; height:44px; background:rgba(255,255,255,.15); border-radius:12px; display:grid; place-items:center; }
43   .brand-name { font-family:'Syne',sans-serif; font-weight:800; font-size:1.3rem; color:#fff; }
44   .left h2 { font-family:'Syne',sans-serif; font-size:2rem; font-weight:800; color:#fff; line-height:1.2; margin-bottom:16px; }
45   .left p { color:rgba(255,255,255,.65); line-height:1.7; font-size:.95rem; margin-bottom:40px; }
```

index.html

```
01 <!DOCTYPE html>
02 <html lang="en">
03 <head>
04 <meta charset="UTF-8">
05 <meta name="viewport" content="width=device-width, initial-scale=1.0">
06 <title>EcoCollect – Smart Waste, Greener Future</title>
07 <link href="https://fonts.googleapis.com/css2?family=Syne:wght@400;600;700;800&family=DM+Sans:ital,opsz,wght@0,9..40,300;0,9..4
08 <style>
09 :root {
10   --green-900: #0d2b1a;
11   --green-800: #1b4332;
12   --green-700: #2d6a4f;
13   --green-500: #40916c;
14   --green-300: #74c69d;
15   --green-100: #d8f3dc;
16   --amber: #e76f51;
17   --amber-lt: #f4a261;
18   --cream: #fafaf7;
19   --white: #ffffff;
20   --text-dark: #1a2e1f;
21   --text-mid: #3a5a40;
22   --text-soft: #6b8f71;
23   --r:8px; --r-lg:16px; --r-xl:24px;
24   --shadow: 0 4px 24px rgba(0,0,0,.08);
25   --shadow-lg: 0 12px 48px rgba(0,0,0,.14);
26 }
27 *, ::before, ::after { box-sizing: border-box; margin: 0; padding: 0; }
28 html { scroll-behavior: smooth; }
29 body { font-family: 'DM Sans', sans-serif; background: var(--cream); color: var(--text-dark); overflow-x: hidden; }
30
31 /* — NAV — */
32 nav {
33   position: sticky; top: 0; z-index: 100;
34   display: flex; align-items: center; justify-content: space-between;
35   padding: 0 5%; height: 70px;
36   background: rgba(250,250,247,.92); backdrop-filter: blur(12px);
37   border-bottom: 1px solid rgba(64,145,108,.12);
38   transition: box-shadow .3s;
39 }
40 nav.scrolled { box-shadow: 0 2px 20px rgba(0,0,0,.08); }
41 .nav-logo { display:flex; align-items:center; gap:10px; text-decoration:none; }
42 .nav-logo-icon { width:36px; height:36px; background:var(--green-800); border-radius:10px; display:grid; place-items:center;
43 .nav-logo-text { font-family:'Syne',sans-serif; font-weight:800; font-size:1.2rem; color:var(--green-800); letter-spacing:-.5
44 .nav-links { display:flex; gap:32px; list-style:none; }
45 .nav-links a { text-decoration:none; color:var(--text-mid); font-size:.9rem; font-weight:500; transition:color .2s; }
```

Chapter 5

Results and Analysis

User Testing & Feedback

- **Participants:** 10 students and 3 local residents.

Quantitative Results

- Pickup request process completed within 2–3 minutes.
- User registration and login worked successfully for all test users.
- Reward points were updated correctly after waste collection.
- The system handled pickup scheduling without data errors during testing.

Qualitative Feedback

- **Students:** “The platform is simple and easy to use.”
- **Users:** “Doorstep pickup makes recycling more convenient.”
- **Test Participants:** “The reward system motivates regular recycling.”
- **Project Review Feedback:** “The application provides an effective solution for plastic waste management.”

For Software Project

Profile Settings

TS

teju shrama

teju0939@gmail.com

Individual

Edit Profile

Full Name

teju shrama

Phone

7019269561

Address

Your address

Area / Locality

BTM Layout

Save Changes

TS

teju shrama

Individual

[Sign Out](#)

My Collector

TS

Rajun Kumar

(ECO-KA-Q042)

★★★★★ 4.8 / 5.0

1,247

Total Collections

96%

On-Time Rate

3 Yrs

Experience

Vehicle

Tata Ace (KA-01-AB-1234)

Area

BTM Layout, Koramangala, HSR

Contact

+91 98765 43210

Schedule

Monday

9:00 AM - 1:00 PM

Wednesday

9:00 AM - 1:00 PM

Friday

9:00 AM - 1:00 PM

Saturday

8:00 AM - 12:00 PM

TS

teju shrama

Individual

[Sign Out](#)

Points & Rewards

0 pts

Total Points Earned

From Collections

Redeemed

Available

Points Log

DATE	DESCRIPTION	POINTS
No points yet. Complete a pickup to start earning!		

TS

teju shrama

Individual

[Sign Out](#)

My Requests

My Pickup Requests

DATE	MATERIALS	WEIGHT	POINTS	STATUS	COLLECTOR
No requests yet. Schedule your first pickup!					

TS

teju shrama

Individual

[Sign Out](#)

Schedule Pickup

Schedule a Pickup

Select Materials *

Plastic

20 pts/kg

Paper & Cardboard

10 pts/kg

Metal & Scrap

25 pts/kg

Glass

8 pts/kg

E-Waste

50 pts/kg

Used Oil

15 pts/L

Old Clothes

5 pts/kg

Organic Waste

3 pts/kg

Preferred Date *

dd-mm-yyyy

Time Slot *

Select time slot...

Estimated Weight (kg)

e.g. 5

Special Instructions

e.g. Call before arriving

Schedule Pickup

A confirmed slot will be assigned within 2 hours.

TS

teju shrama

Individual

[Sign Out](#)

Chapter 6

Conclusion & Future Work

Conclusion

The Smart Plastic Recycling: A Reward-Based Solution successfully improves the process of plastic waste management through a simple and user-friendly digital platform. The system encourages people to recycle plastic waste by providing doorstep pickup services and reward points for participation. It reduces the inconvenience of traditional recycling methods and promotes environmental awareness among users. The platform also provides secure user management, efficient pickup scheduling, and reliable data storage using modern web technologies. Overall, the project supports sustainable waste management practices and contributes to reducing plastic pollution in the community.

Future work

Mobile application development for easier access.

GPS-based live tracking for waste collection.

AI-based waste classification and segregation.

Online reward redemption and payment integration.

Expansion of the system to support multiple recyclable waste categories.

References

The project idea, system design, and implementation concept are original. AI tools were used only for development assistance, coding support, and documentation guidance.

References

OpenAI – ChatGPT

Used for coding assistance, content generation, debugging, and project documentation support.

Anthropic – Claude AI

Used for idea refinement, frontend suggestions, and development guidance.

Firebase Documentation

Referred for authentication, database, and hosting implementation.

MDN Web Docs

Used for HTML, CSS, and JavaScript references.

React Documentation

Referred for frontend development concepts and components.